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SURG-27. 17-YEAR EXPERIENCE OF SURGICAL MANAGEMENT OF INTRADURAL EXTRAMEDULLARY (IDEM) TUMORS: CLINICAL OUTCOMES FROM A LOW- AND MIDDLE-INCOME COUNTRY
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INTRODUCTION: Surgical outcomes of intradural extramedullary (IDEM) tumors have been well-reported in the literature from high-income countries. However, limited contemporaneous data from low- and middle-income countries (LMICs) exists. This study sought to analyze the clinical outcomes of all IDEM tumor cases operated during a 17-year period at a resource-limited setting in an LMIC. METHODS: The present work was conducted and reported in accordance with Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines. A retrospective cohort study was carried out to analyze all IDEM tumor cases, operated upon by a consultant neurosurgeon, during 2004-2021, at a resource-limited setting in Southern India. All patients underwent routine biochemical and radiological investigations, including contrast-enhanced MRI. Laminectomy and tumor resection under the microscope were done without using intraoperative neuro-monitoring. Nerve fibers running through the tumor were sacrificed. Bilateral facet joints were preserved to ensure integrity without spinal fusion. All cases received a dural closure with 5-0 Prolene, with postoperative MRI confirming complete resection. Patients received telephonic and/or in-person follow-up visits. RESULTS: 50 patients with a mean age of 46.3 years (range 13-80) were included, with 31 males and 19 females. Their clinical presentation included radiculopathy (78%), radiculomyelopathy (12%), and radiculomyelopathy with bowel and bladder symptoms (10%). Histologically, schwannomas were the most common (52%), followed by neurofibromas (34%), meningiomas (12%), and ependymomas (2%). Two individuals were diagnosed with Neurofibromatosis Type 1. All meningiomas occurred in the thoracic region, with 83.3% of cases being in females. All patients had symptomatic relief postoperatively. No major adverse events occurred, including CSF leaks, permanent neurological deficits, or spinal instability. Recurrence occurred in two 'dumbbell' tumors (4%) at 5 and 7 years after primary excision during follow-up, managed with successful re-resection. CONCLUSION: Microsurgical excision of IDEM tumors carries high safety and efficacy, including in resource-limited settings like LMICs.